

MODEL QUESTION PAPER

TED(10) 5042

(Revision _2010)

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY

CONTROL SYSTEM

(Maximum marks :100)

Time : 3 Hours

PART – A

(Answer all questions . Each question carries 2 marks)

- I
1. Why is negative feed back preferred in a closed loop system?
 2. State the differentiation theorem in Laplace transform ?
 3. Define node in Signal flow graph ?
 4. Define gain cross over frequency ?
 5. If there is no change in the first column of Routh array , what it indicates ?

(5 X 2 = 10)

PART – B

(Answer any 5 questions .Each question carries 6 marks)

- II
1. How is the mathematical model of a physical system obtained. Give an example for it?
 2. Find the Laplace transform of Unit Step Function?
 3. Derive the transfer function of mechanical translational system?
 4. What is a signal flow graph? What are the properties of signal flow graph?
 5. Obtain the time response of first order control system to unit step input function?
 6. Draw the Bode plot for $G(s)=K$?
 7. Define stability of a control system? Differentiate absolute and relative stability?

(5 X 6 = 30)

PART C

(Answer one full question from each unit. Each question carries 15 marks)

MODULE I

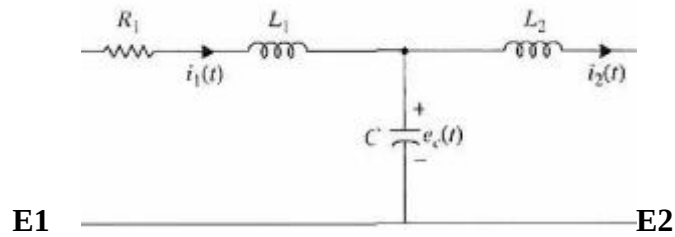
- III
- a). Explain linear time invariant and linear time varying system? 8
 - b) A functional Laplace domain is given by $F(S) = 2/S - 1/(S+3)$. Obtain its final value using final value theorem? 7

OR

- IV
- a) Obtain the Laplace inverse of $F(S) = S+2 / (S^2 - 5S + 6)$ 8
 - b) State and prove Initial value theorem? 7

MODULE II

- V
- a) Obtain the transfer function of the network shown in figure. 10



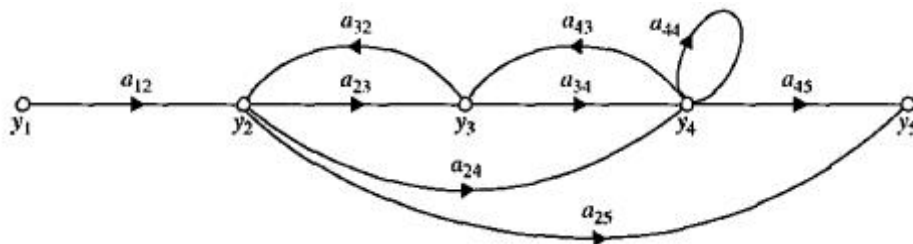
b) Write the procedure for finding the transfer function of a control system ?

5

OR

VI a) A signal flow graph is shown in figure. Determine the overall transmittance relating to y_1 and y_5 ?

10



b) Write Mason's gain formula? Explain each term?

5

MODULE III

VII a) Derive the time response of second order control system to unit step input function?

10

b) Write short notes on different test signals?

5

OR

VIII a) Describe the static error coefficients K_p , K_v , K_a ?

8

b) Explain Type 0, Type 1 and Type 2 control systems?

7

MODULE IV

IX Using Routh stability criterion determine the stability of the system represented by the characteristic equation $S^5 + S^4 + 2S^3 + 2S^2 + 3S + 5 = 0$. Comment on the location of the roots of the characteristic equation?

15

OR

X Construct root locus for a control system whose open loop transfer function is given by $G(S) = K / S(S+6)$ and $H(S) = 1$.

15

(4X15 =60)
